Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

FIFTH SEMESTER BTECH. (E & C) DEGREE END SEMESTER EXAMINATION
FEBRUARY-MARCH 2022
SUBJECT: INTRODUCTION TO COMMUNICATION SYSTEMS (ECE -4304)

TIME: 75 minutes **MAX. MARKS: 20**

Instructions to candidates

- Answer **ALL** questions.
- Missing data may be suitably assumed.

Q. No.	Questions	Marks
1A.	With neat block diagrams, explain modulation and demodulation process used in electronic communication system	5
1B.	With neat diagrams, explain the operation of cellular mobile communication system	3
1C.	Explain the PIN photodiode with necessary diagrams	2
2A.	Explain the power budget. Assume 10km fiber optic link is to be installed, connected at both ends with connectors and following datas are available (i) 20 sections of fiber optic cable each length 500 meter with an attenuation of 1dB/km (ii) Connector loss of 1.3 dB/ connector (iii) Splice loss of 0.5 dB/splice (iv) Receiver sensitivity -50 dBm (v) System margin 5 dB	5
2B	Find the core radius necessary for single mode operation at 1320nm of a step-index fiber with n_1=1.48 and n_2 = 1.478. What are the numerical aperture and maximum acceptance angle of this fiber?	3
2C	A pulsed radar has a pulse repetition rate 1700 pulses/second and transmits rectangular pulses of duration 15μS. What is the maximum range can a target have if no part of its delayed pulse is to overlap any part of a transmitted pulse and not be delayed more than one PRF interval?	2